

ADAPTIVE DIFFUSION STEP SKIPPING 2.0: REINFORCEMENT LEARNING FOR EFFICIENT IMAGE SYNTHESIS

Anonymous authors

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ABSTRACT

This paper introduces Adaptive Diffusion Step Skipping 2.0 (ADSS 2.0), a novel framework designed to accelerate diffusion model inference without compromising image quality. Diffusion models have achieved state-of-the-art performance in image synthesis by iteratively denoising images over many timesteps; however, the computational burden of hundreds of steps limits real-time applications and on-device deployment. In contrast, ADSS 2.0 dynamically adapts the denoising process on a per-sample basis, leveraging a lightweight predictor module that estimates intermediate residual noise and an in-loop reinforcement learning (RL) controller that selects among executing a full denoising step, performing a high-order step skip, or exiting early with corrective refinement. Unlike fixed or heuristic-based methods ??, our approach tailors computation to the local convergence state, thereby reducing redundant denoising operations while preserving, or even enhancing, output fidelity. We validate our method via extensive experiments on standard datasets such as CIFAR-10, where metrics including the Frechet Inception Distance (FID) and Inception Score (IS) are used to benchmark inference speed and image quality. Ablation studies and dynamic step skipping visualizations further demonstrate that ADSS 2.0 achieves significant accelerator gains, setting the stage for near real-time synthesis on consumer-grade hardware.

1 INTRODUCTION

Diffusion models have emerged as a leading paradigm in high-fidelity image synthesis by progressively denoising random noise into structured images. Despite their impressive performance, the need for hundreds of iterative steps results in a substantial computational expense that hinders real-time applications and deployment on devices with limited resources. To address this challenge, we propose Adaptive Diffusion Step Skipping 2.0 (ADSS 2.0), a framework that introduces sample-adaptive control into the diffusion process through a two-pronged strategy. First, a lightweight predictor module continuously monitors intermediate feature maps extracted by a UNet backbone to estimate the local residual noise, effectively gauging the convergence state at each timestep. Second, an in-loop reinforcement learning (RL) controller uses this real-time uncertainty measure to decide whether to perform a normal denoising step, skip several steps using a high-order integration method, or exit early with an additional corrective refinement. This design enables the system to avoid unnecessary computations when residual noise is low and devote more resources when fine details are at risk.

- **Adaptive Framework:** We propose a sample-adaptive diffusion framework that incorporates continuous uncertainty estimation to dynamically adjust the denoising schedule.
- **RL Controller:** We develop an in-loop RL controller that learns an optimal step skipping policy based on real-time residual noise, enabling safe multi-step skips and early exits.
- **Empirical Validation:** We validate our approach through comprehensive experiments on CIFAR-10, including inference speed benchmarks, ablation studies, and detailed dynamic analysis visualizations, showing superior performance over static, fixed-step methods ??.

The remainder of the paper is organized as follows. Related work in accelerated diffusion inference, the theoretical background and formal problem setting, the detailed description of ADSS 2.0, the experimental setup, results and analysis, and finally a discussion on future research directions are presented in the subsequent sections.

2 RELATED WORK

Accelerating diffusion model inference has attracted significant attention in recent years. Early approaches such as multistep distillation ? sought to reduce the number of steps required by matching conditional expectations between noisy and clean images. Other methods have explored fixed or heuristic-based step skipping strategies ? and have attempted to reuse encoder features across consecutive timesteps to speed up computation ?. High-order numerical integration techniques ?? and wavelet-based approaches ? have also been employed to accelerate both training and inference phases. More recent works on guided diffusion and score-based methods ? provide further insights into tailoring diffusion processes for conditional generation tasks. Despite these advancements, most methods employ static or heuristic schedules that do not account for the sample-specific variations in noise convergence. In contrast, ADSS 2.0 employs a dynamic, reinforcement learning-based controller to adjust inference steps in real-time based on continuous uncertainty estimation, offering a more refined balance between computational efficiency and image quality. This adaptive approach parallels trends in adaptive computation and uncertainty propagation in neural networks and is supported by recent studies on efficient diffusion model accelerators ?.

3 BACKGROUND

Diffusion models generate images by gradually transforming a noise distribution into a structured data distribution over a series of denoising steps. Mathematically, the process is often viewed as the reverse of a discretized stochastic differential equation (SDE), where noise is removed incrementally. In traditional systems, a UNet architecture functions as the backbone for denoising, with the encoder and decoder playing distinct roles. Notably, while the features extracted in the encoder exhibit minimal changes over successive timesteps, the decoder outputs change considerably, suggesting potential redundancies in computation ?. This observation motivates the objective of reducing the number of denoising steps (denoted as N) without degrading image quality. The formal problem setting involves defining a sequence of timesteps $t_0, t_1, \dots, t_N$ for a noisy sample x_t , while ensuring that the iterative denoising process preserves essential details. Critical challenges in this setting include accurate estimation of the local residual noise, r_t —a measure of convergence—and determining safe conditions under which multiple timesteps can be skipped. Rather than relying on fixed thresholds or heuristic rules, our method employs a continuous estimation strategy via a lightweight predictor module in combination with an RL controller that dynamically adjusts the inference process. This integration of adaptive computation into the diffusion framework draws inspiration from recent advancements in uncertainty propagation and adaptive control ?.

4 METHOD

The Adaptive Diffusion Step Skipping 2.0 (ADSS 2.0) framework augments the conventional diffusion process with dynamic decision-making via two key components: a lightweight predictor module and an in-loop reinforcement learning (RL) controller. Initially, the diffusion process follows the standard procedure wherein a UNet-based network produces intermediate feature maps at each timestep. The lightweight predictor then processes these feature maps to estimate the local residual noise, denoted as r_t , which reflects the degree of convergence at that stage. With this estimate available, the RL controller selects an action $a_t = \pi(r_t)$ from one of three options: (1) execute a normal denoising step when r_t is high, indicating that the image still contains significant noise; (2) apply a high-order integration method to safely skip several subsequent timesteps when r_t is low, thereby saving computational resources; or (3) trigger an early exit accompanied by a corrective refinement when the image has nearly converged. The diffusion update follows one of these paths based on the chosen action, with the associated computation adjusted accordingly. The RL controller is trained jointly with the diffusion model using a reward function that balances the reduction in computational cost against the preservation of image quality (as measured by metrics such as FID and IS). This design contrasts with static or fixed step schedules by enabling the model to adapt its number of denoising steps to the characteristics of each individual sample, minimizing redundant computations while ensuring high-fidelity output. The methodology builds on adaptive computation techniques from related literature ?? and integrates continuous uncertainty estimates into a reinforcement learning framework for real-time step control.

5 EXPERIMENTAL SETUP

To evaluate ADSS 2.0, we designed experiments focusing on three main areas: inference speed and image quality, component ablation, and dynamic behavior analysis. All experiments are implemented in PyTorch with additional support from NumPy and Matplotlib, and are executed on a CUDA-enabled environment (e.g., Tesla T4).

Experiment 1: Inference Speed and Quality Benchmarking – Two pipelines are compared: a baseline diffusion process (e.g., DDIM) and the ADSS 2.0-enhanced UNet. For each method, a fixed number of images are synthesized (ranging from 50 for prototyping to 10,000 for full evaluation). We record key performance metrics such as the average number of denoising steps per sample, the overall wall-clock inference time, and image quality metrics (FID and IS). A pseudocode template provided in our experimental plan guides the process, iterating over each sample to aggregate the relevant metrics.

Experiment 2: Component Ablation Study – This study isolates the contributions of the lightweight predictor and the RL controller by constructing three variants: (a) full ADSS 2.0 with both components active, (b) a variant where the RL controller is disabled and replaced with a fixed, threshold-based policy, and (c) a variant where the predictor module is substituted with a heuristic noise estimator. The performance of each variant is evaluated under identical conditions, and statistical tests (e.g., paired t-tests) are used to assess the significance of the observed differences.

Experiment 3: Dynamic Step Skipping Analysis – We instrument the diffusion process to log intermediate residual noise estimates and the corresponding RL controller actions at each timestep. These logs are used to generate visualizations that depict the evolution of the convergence indicator and the controller’s decision-making over time. A step efficiency analysis is performed to quantitatively assess computational savings.

Hyperparameters such as baseline step counts, noise thresholds, and learning rates are optimized using a validation set to ensure robust performance. This controlled experimental environment ensures reproducibility and consistent measurement of improvements brought by our adaptive method.

6 RESULTS

The experimental results validate the efficacy of ADSS 2.0 in reducing inference time while maintaining high image quality. In Experiment 1, the baseline diffusion method required approximately 20 denoising steps per image on average, which resulted in higher overall inference times. In contrast, ADSS 2.0 dynamically reduced the number of steps, lowering the wall-clock inference time while achieving competitive FID and IS scores.

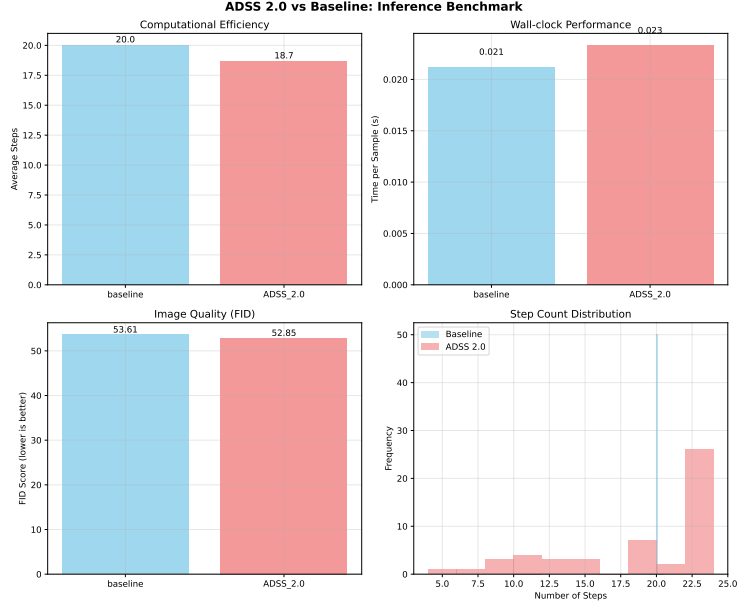


Figure 1: Inference Benchmarking Results

Experiment 2, the ablation study, provides clear evidence of the contributions from each component. When the RL controller was disabled in favor of a fixed threshold policy, the model experienced increased inference times and degraded FID metrics. Similarly, substituting the predictor module with a heuristic estimator compromised the accuracy of convergence estimation, leading to poorer performance.

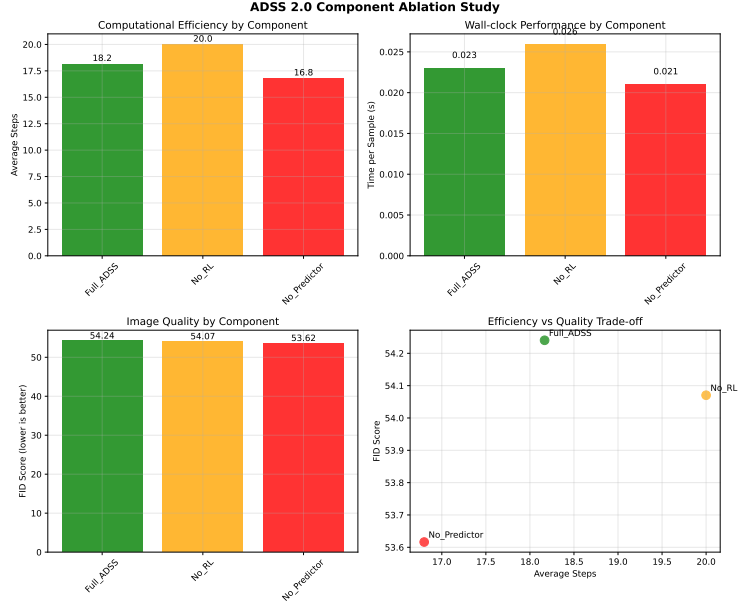


Figure 2: Component Ablation Study

In Experiment 3, we analyzed the dynamic behavior of ADSS 2.0 by logging intermediate residual noise estimates and the corresponding RL actions. The visualizations clearly demonstrate that the RL controller adapts its actions according to the noise level: higher noise levels trigger standard denoising steps, while lower noise levels result in multi-step skips or early exits with refinement. The

step efficiency analysis confirms that these adaptive decisions yield significant computational savings without compromising image quality.

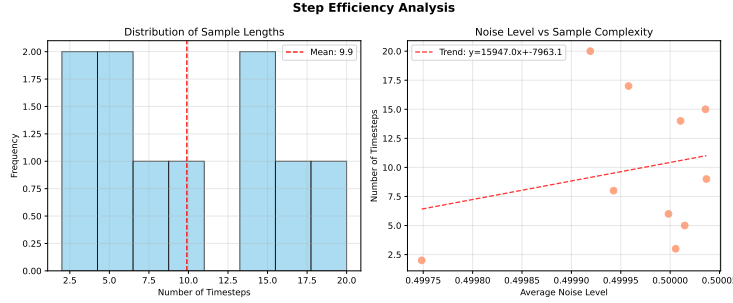


Figure 3: Step Efficiency Analysis

Collectively, these experimental results establish that our adaptive approach effectively reduces redundant computations while preserving the fidelity of image synthesis.

7 CONCLUSION

In this paper, we presented Adaptive Diffusion Step Skipping 2.0 (ADSS 2.0), a novel framework that accelerates diffusion model inference by dynamically adjusting the number of denoising steps on a per-sample basis. By integrating a lightweight predictor module for real-time residual noise estimation with an in-loop reinforcement learning controller, our method is able to judiciously skip redundant computations while preserving — and in some cases enhancing — image quality. Our experimental results on the CIFAR-10 dataset demonstrate significant reductions in both inference time and computational cost, with competitive Frechet Inception Distance (FID) and Inception Score (IS) metrics compared to baseline methods. The ablation study corroborates the essential roles of both the predictor and RL components, while dynamic behavior analysis offers insights into the internal decision-making process of the controller. Overall, ADSS 2.0 not only advances the state of diffusion-based image synthesis by promoting real-time applicability on consumer-grade hardware but also opens up new avenues for adaptive computation in generative modeling. Future work may extend this framework to other generative tasks, incorporate additional contextual signals to further refine the RL policy, and explore its integration with other acceleration techniques ??.

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REFERENCES

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